

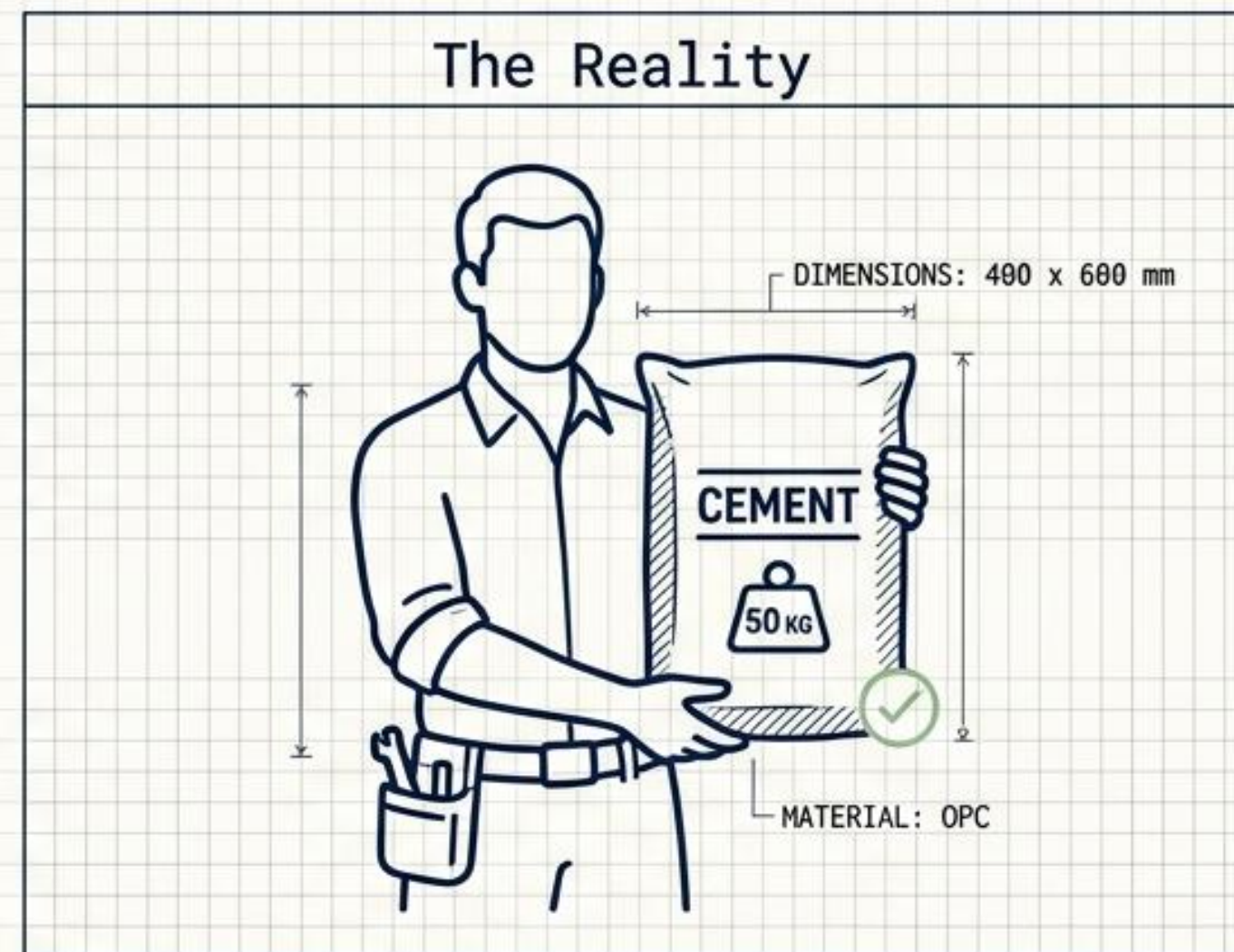
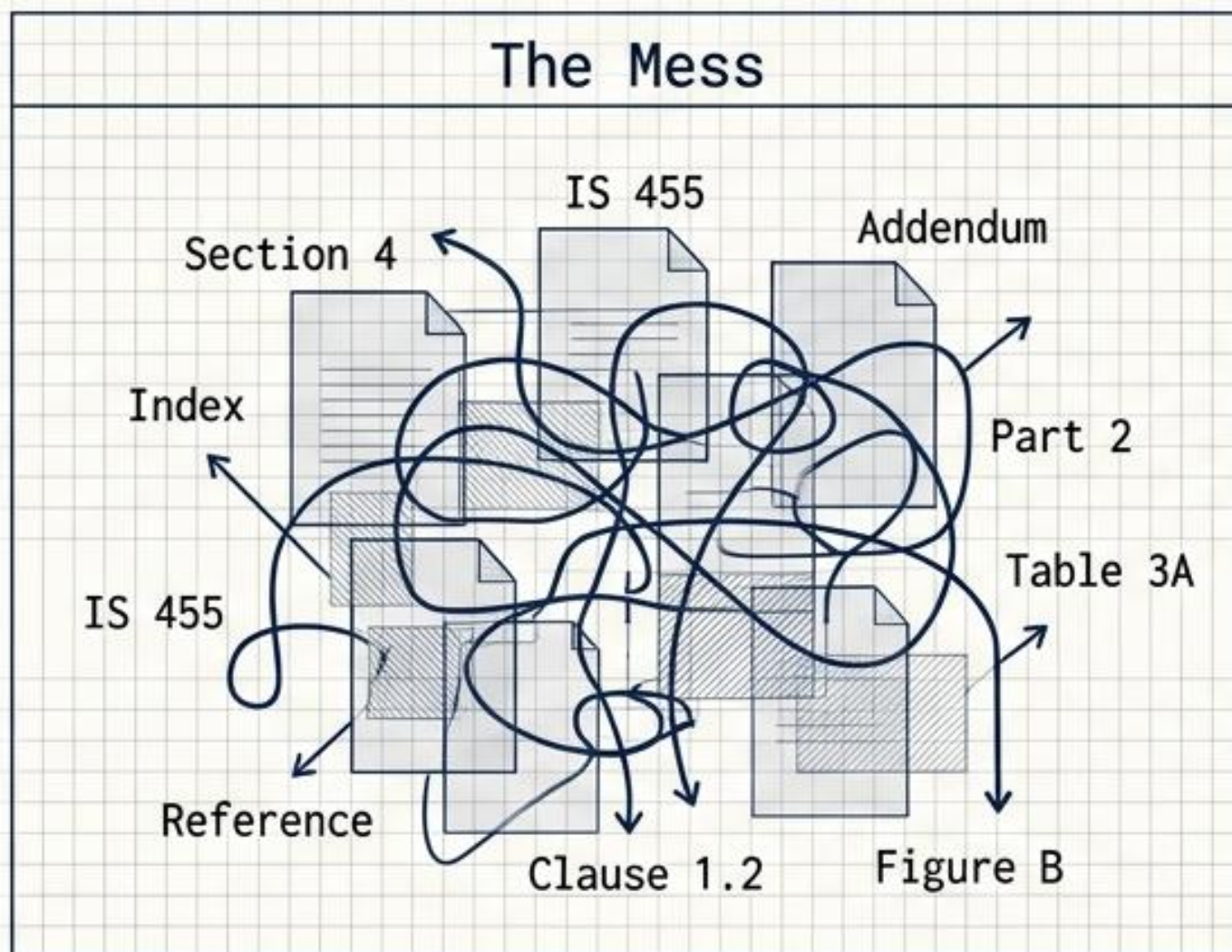


BIS-Compass

Accelerating Standard Discovery for Indian MSEs.

system: recommendation_engine

The Discovery Bottleneck for MSEs



Indian micro and small enterprises spend hours mapping building materials to BIS standards. The challenge isn't just finding a number—it's navigating sectioned PDFs and indexed pages accurately without an engineering degree.

The goal: Turn a manual, error-prone search into an AI-assisted workflow.

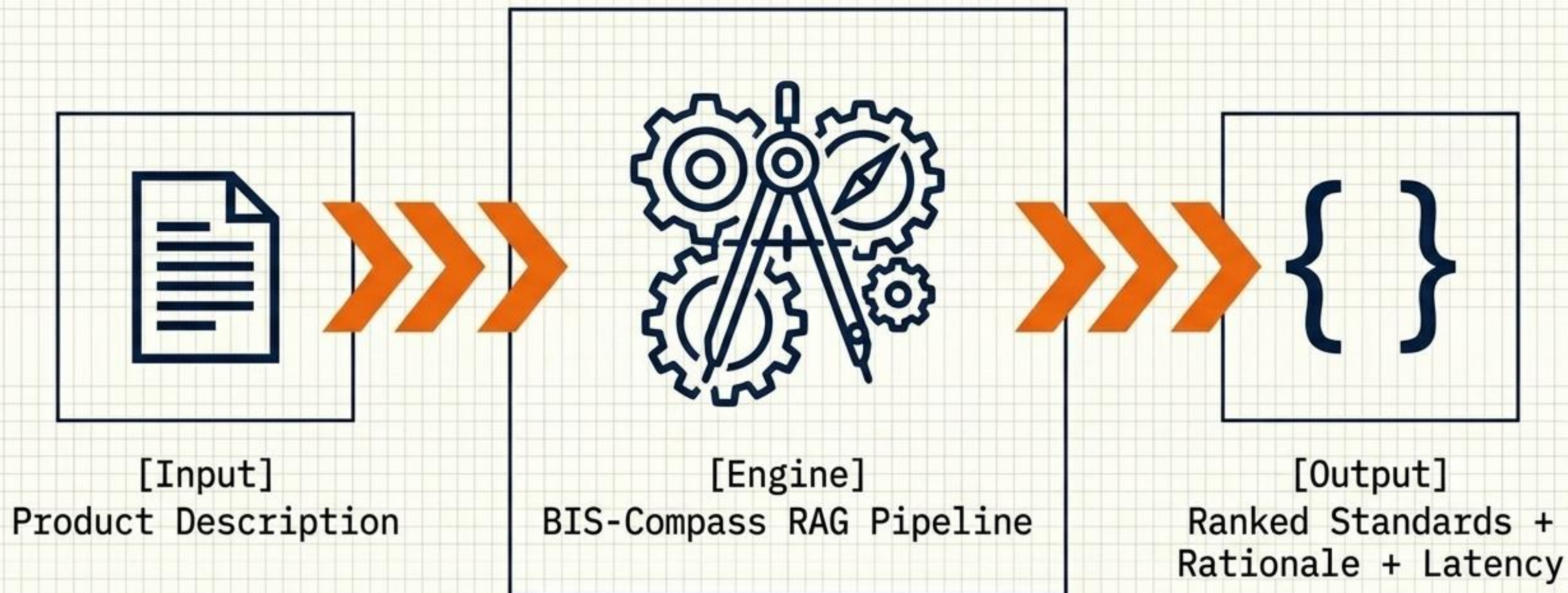
Diagnostic: Manual Search vs. BIS-Compass

Manual Search

BIS-Compass Workflow

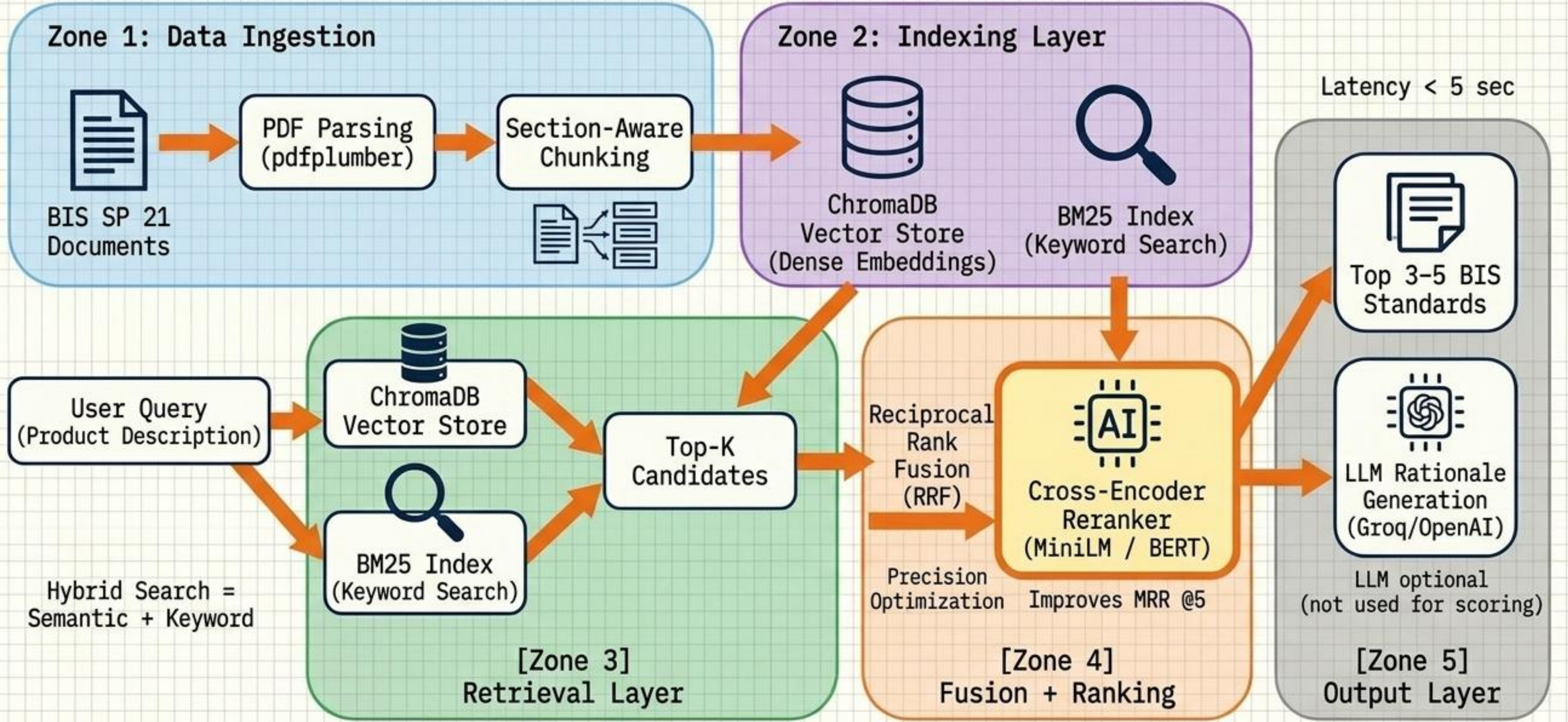
Reading summaries & cross-checking indices	AI-Assisted Recommendation Engine ✓
Hours or Minutes	Under 4 Seconds ✓
Poor (Lost in PDFs)	High (LLM-generated rationales) ✓
Guesswork	Judge-Ready JSON schema ✓

The Core Engine Pathway



Designed strictly for the hackathon evaluator.
Reproducible, transparent, and bound to the required inference.py schema.

End-to-End RAG Architecture



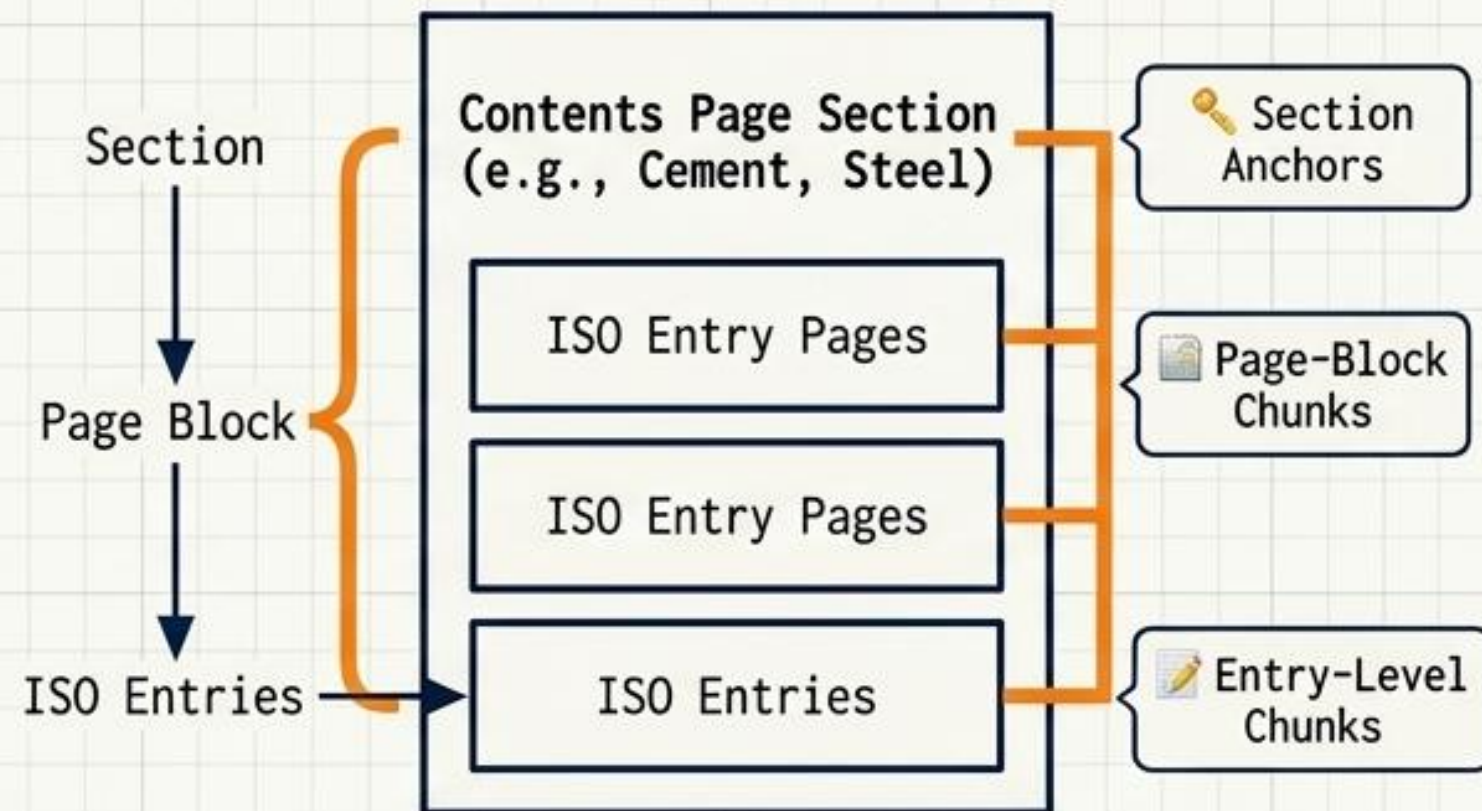
Section-Aware Chunking for Structured PDF Retrieval (BIS Standards)

Naive Chunking Approach



Breaks semantic context
Ignores document structure
Misses standard groupings

Section-Aware Chunking Approach



Preserves human-readable structure
Groups related ISO standards
Enables clean and precise retrieval

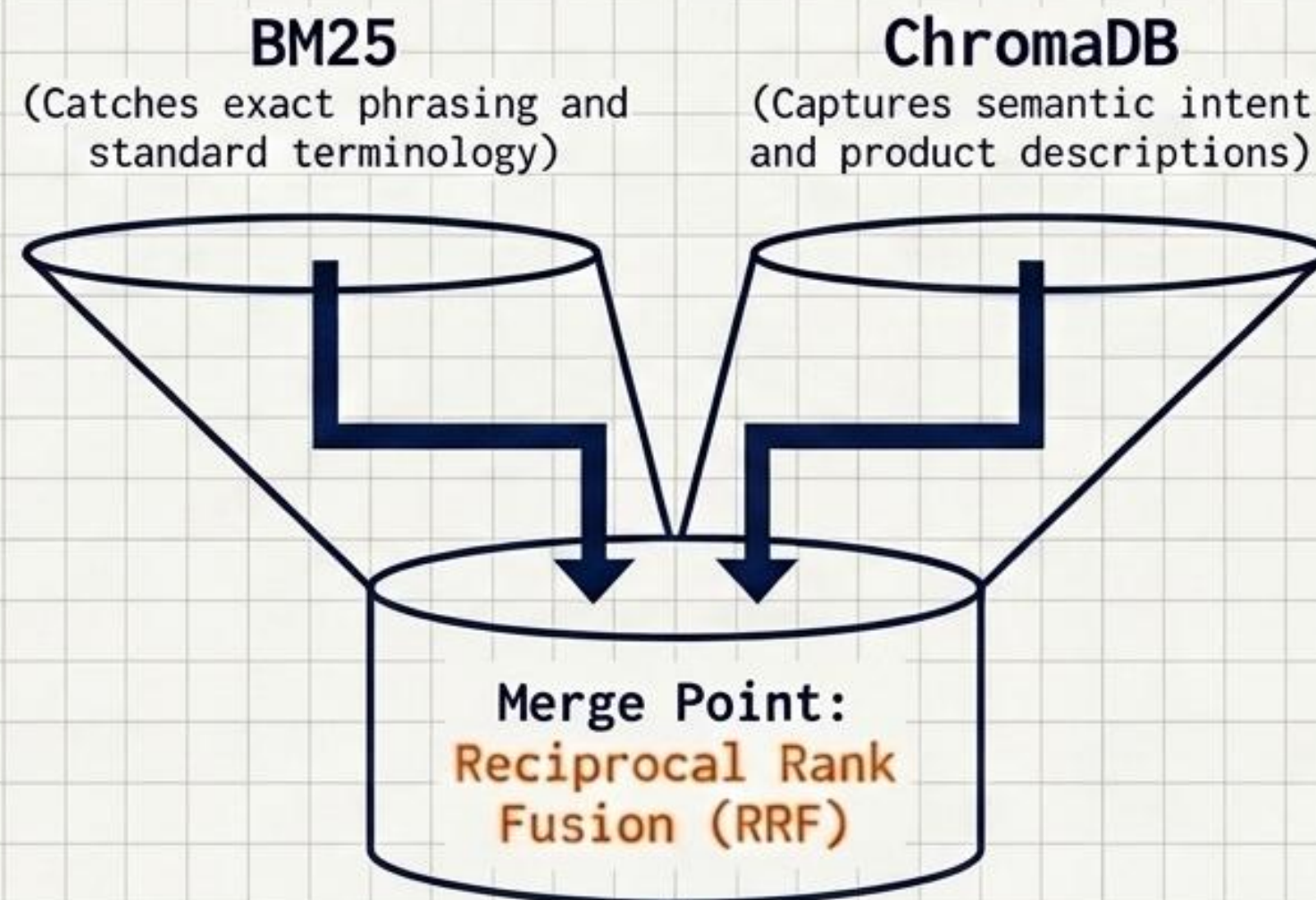
Hybrid Retrieval Strategy

 **BM25** → exact keyword / ISO standard match

 **Vector DB (ChromaDB)** → semantic similarity

 **Combine both** → improved precision + recall

Fusing Semantic and Exact Retrieval



Instead of choosing between keyword or semantic search, we fuse both rankings to get the best of both methods

Cross-Encoder Filter: Acts as a final quality filter, ensuring the most relevant standard appears at the top.

Demo highlights (Input)

BIS-COMPASS

Intelligent Standards Navigator for Indian MSEs

PRODUCT DESCRIPTION

I need to comply with the regulations for coarse and fine aggregates derived from natural sources intended for use in structural concrete

TRY THESE EXAMPLES:

OPC Cement 53 Grade

TMT Steel Bars Fe500

Coarse Aggregate 20mm

Fly Ash Bricks

Hollow Concrete Blocks

SCAN STANDARDS →

5 standards found

⚡ 0.00s

STANDARDS FOUND

5

QUERY TIME

0.00s

HIGH RELEVANCE

0

Demo Highlights (Output)

01 IS 383: 1970

MEDIUM

Specification for Coarse and Fine Aggregates from Natural Sources for Concrete

Aggregates

Retrieved from the BIS aggregates corpus for query matching.

Score:
0.00

02 IS 383: 2016

MEDIUM

Coarse and Fine Aggregate for Concrete - Specification (Revised)

Aggregates

Retrieved from the BIS aggregates corpus for query matching.

Score:
0.00

03 IS 458: 2003

MEDIUM

Specification for Precast Concrete Pipes (with and without Reinforcement)

Concrete

Retrieved from the BIS concrete corpus for query matching.

Score:

Public Test Set Performance

100.00%

Hit Rate @3

1.0000

MRR @5

3.46 sec

Avg Latency

Validated strictly against the official `eval_script.py`.
The first query may be slower on a cold start because it warms up models and loads the retrieval stack, so we keep that warmup outside the measured loop

Translating Metrics into MSE Impact

100% Hit Rate



Unshakable Confidence

Non-experts can trust the compliance mapping without second-guessing.

3.46s Latency



Accelerated Workflows

Manual searches that took hours are reduced to seconds.

Generated Rationales



Complete Traceability

Users don't just receive a raw prediction; they get the *why* required for compliance files.

Team & Acknowledgements

Tech Stack

Python

FastAPI

React

ChromaDB

sentence-transformers

BM25

Cross-encoder

Data Source

BIS SP 21 building-material standards summary.

Roadmap

Tighter section-page mapping, stronger reranking calibration, richer reviewer UI.

Team Members

Dhamagatla Rohith Kumar
Jashwanth Reddy
Devansh Dewan

Turning a dense standards PDF into a fast, transparent, judge-ready recommendation engine for Indian MSEs.